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Systemic Pathology



Pathology of Reproductive System

Sexually Transmitted Diseases (STDs)

- STIs have a profound impact on sexual and reproductive health worldwide.
- More than 1 million STIs are acquired every day, 374 million new infections with 1 of 4 STIs: chlamydia (129 million), gonorrhoea (82 million), syphilis (7.1 million) and trichomoniasis (156 million).
- More than 490 million people were estimated to be living with genital herpes in 2016
- It is estimated 300 million women have an HPV infection, the primary cause of cervical cancer and anal cancer among men who have sex with men.
- An estimated 296 million people are living with chronic hepatitis B globally.

- STIs can have serious consequences beyond the immediate impact of the infection itself.
 - STIs like herpes, gonorrhoea and syphilis can increase the risk of HIV acquisition.
 - Mother-to-child transmission of STIs can result in stillbirth, neonatal death, low-birth weight and prematurity, sepsis, neonatal conjunctivitis and congenital deformities.
 - HPV infection causes cervical and other cancers.
 - Hepatitis B resulted in an estimated 820 000 deaths in 2019, mostly from cirrhosis and hepatocellular carcinoma.
 - STIs such as gonorrhoea and chlamydia are major causes of pelvic inflammatory disease and infertility in women.

Common STDs

- More than 30 different bacteria, viruses and parasites are known to be transmitted through **sexual contact**, including vaginal, anal and oral sex.
- Some STIs can also be transmitted from **mother-to-child** during pregnancy, childbirth and breastfeeding.
- **Eight pathogens** are linked to the greatest incidence of STIs.
 - Of these, 4 are currently curable: syphilis, gonorrhoea, chlamydia and trichomoniasis.
 - Other 4 are incurable viral infections: hepatitis B, herpes simplex virus (HSV), HIV and human papillomavirus (HPV).
- In addition, emerging outbreaks of new infections that can be acquired by sexual contact such as monkeypox, *Shigella sonnei*, *Neisseria meningitidis*, Ebola and Zika, as well as re-emergence of neglected STIs such as lymphogranuloma venereum.
- These herald increasing challenges in the provision of adequate services for STIs prevention and control.

Symptoms of STDs

Asymptomatic

Symptoms such as

- Sores or bumps on the genitals or in the oral or rectal area.

- Painful or burning urination

- Discharge from the penis

- Unusual or odorous vaginal discharge

- Unusual vaginal bleeding

- Pain during sex

- Sore, swollen lymph nodes, particularly in the groin but sometimes more widespread

- Lower abdominal pain

- Fever

- Rash over the trunk, hands or feet

Prevention

- Condoms offer one of the most effective methods of protection against STIs, including HIV. Although highly effective, condoms do not offer protection for STIs that cause extra-genital ulcers (i.e., syphilis or genital herpes). When possible, condoms should be used in all vaginal and anal sex.
- Safe and highly effective vaccines are available for 2 viral STIs: hepatitis B and HPV.
- Adult voluntary medical male circumcision, microbicides, and partner treatment.

Testicular Atrophy

- Testis is smaller than its normal size. It can be seen unilateral or bilateral.
- Causes
 - **Orchitis Eg:** mumps
 - **Varicocele:** Varicocele is the enlargement of testicular veins (testicular veins)
 - **Testicular torsion or thrombosis:** spermatic cord, in which the testicular vessels are located, rotates around itself and as a result, insufficient blood flow to the testis is not reached.
 - **Old age:** Due to aging, testicular sizes decrease in men. Testicular atrophy due to aging is bilateral
 - **Excessive alcohol use:** Due to reduce serum testosterone hormone. Alcohol-induced testicular atrophy (testicular atrophy) is bilateral.
 - **Receiving external testosterone therapy:**
 - **Estrogen or anabolic steroid use**
 - **Trauma**
 - **Undescended testis (cryptorchidism)**

Testicular Atrophy - Pathology



- Complications
- **Infertility**
- **Erection problem**
- **Testicular cancer**
- **Psychological problems**

Prostate Hyperplasia

- Common condition encountered in aging men and a common cause of lower urinary tract symptoms
- Nonmalignant growth or hyperplasia of prostate tissue
- Characterized by stromal and epithelial cell proliferation in the prostate transition zone, which surrounds the urethra. This leads to urethral compression and the development of bladder outflow obstruction, which can result in clinical manifestations of LUTS, urinary retention, or infections due to incomplete bladder emptying

- BPH arises due to the loss of homeostasis between prostatic cellular proliferation and apoptosis or cell death. This imbalance favors cellular proliferation without intervention. The result is increased numbers of prostatic periurethral epithelial and stromal cells, which can be seen histopathologically.
- Risk factors
 - Diabetes, diet (Beta-carotene, carotenoids, and vitamin A seem somewhat protective, while excessive alcohol ingestion, heavy caffeine intake, and high-dose supplemental vitamin C tend to increase BPH risk and symptoms), genetic factors, localized inflammation (obesity, low-grade or chronic prostatitis, compression of the prostatic ducts, and autoimmune disorders), obesity, and metabolic syndrome (refers to conditions that include hypertension, glucose intolerance/insulin resistance, and dyslipidemia.)
- Clinical symptoms
 - Increased bladder outlet resistance
 - Lower urinary tract symptoms (LUTS), including:
 - Obstructive symptoms (hesitancy, intermittent stream and straining)
 - Urinary bladder irritation symptoms (frequency, urgency and urge incontinence)
 - Urinary retention (whether acute or chronic)

Prostate Hyperplasia



Infertility and Subfertility

- Terms subfertility and infertility are often used interchangeably, but they aren't the same.
- Subfertility is a **delay in conceiving**.
- Infertility is the **inability to conceive naturally after one year of trying**.
- In subfertility, the possibility of conceiving naturally exists, but takes longer than average.

Infertility (WHO Facts)

- Disease of the male or female reproductive system defined by the failure to achieve a pregnancy after 12 months or more of regular unprotected sexual intercourse
- 48 million couples and 186 million individuals live with infertility globally

Causes of Infertility

- Caused by a number of different factors, in either the male or female reproductive systems. However, it is sometimes not possible to explain the causes of infertility.



- **Tubal disorders** such as blocked fallopian tubes, which are in turn caused by untreated sexually transmitted infections (STIs) or complications of unsafe abortion, postpartum sepsis or abdominal/pelvic surgery
- **Uterine disorders** which could be inflammatory in nature (such as endometriosis), congenital in nature (such as septate uterus), or benign in nature (such as fibroids)
- **Disorders of the ovaries**, such as polycystic ovarian syndrome and other follicular disorders
- **Disorders of the endocrine system** causing imbalances of reproductive hormones - pituitary cancers and hypopituitarism.



- **Obstruction of the reproductive tract** - can occur in the tubes that carry semen (such as ejaculatory ducts and seminal vesicles). Blockages are commonly due to injuries or infections of the genital tract.
- **Hormonal disorders** leading to abnormalities in hormones produced by the pituitary gland, hypothalamus and testicles. Hormones such as testosterone regulate sperm production.
- **Testicular failure to produce sperm**- medical treatments that impair sperm-producing cells (such as chemotherapy).
- **Abnormal sperm function and quality**. Conditions or situations that cause abnormal shape (morphology) and movement (motility) of the sperm negatively affect fertility. For example, the use of anabolic steroids can cause abnormal semen parameters such sperm count and shape

Dysfunctional uterine bleeding

- Excessively heavy, prolonged or frequent bleeding of uterine origin that is not due to pregnancy or any recognizable pelvic or systemic disease
- Dysfunctional uterine bleeding is the diagnosis in 40–60% of women with excessive menstrual bleeding which is defined as greater than 80 mL blood loss

- There are two types of DUB: ovulatory and anovulatory.
- Ovulatory DUB accounts for about 80% of cases. In ovulatory DUB the menstruation is regular, preceded by ovulation and heavy but of normal duration. It is most common in women in their 30s.
- Anovulatory DUB is more likely to occur at the beginning and end of the reproductive years. The menstrual cycle is irregular and the bleeding is heavy and may be prolonged
- In women with abnormal uterine bleeding both uterine (endometrial and myometrial) and systemic causes need to be excluded before DUB can be diagnosed

- Assessment
 - Clinical assessment
 - Symptoms and signs that may signify heavy menstrual bleeding include: an unusual increase in blood loss, more than 7 days of bleeding, bleeding or flooding not contained within pads or tampons (particularly if wearing the largest size), clots greater than 3 cm, and, presence of signs of anaemia or iron deficiency on blood testing

QUESTION

- What is the rationale of prescribing Nonsteroidal anti-inflammatory drugs (NSAIDs) and Tranexamic acid for the medical management of DUB.

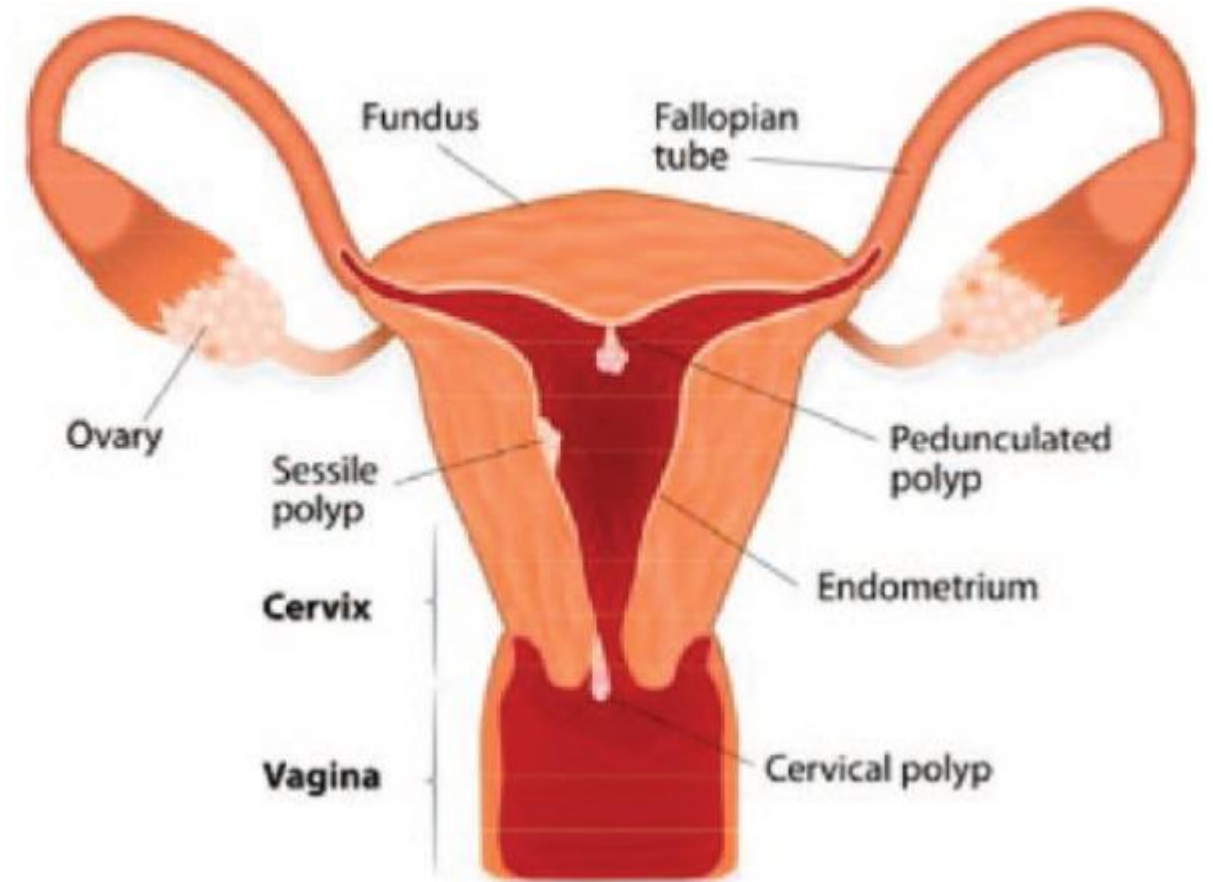
Uterine causes of bleeding

- Endometrial polyps, hyperplasia, and rarely endometrial carcinoma, may cause heavy menstrual bleeding but also may present with intermenstrual bleeding.
- Uterine fibroids, in particular submucous fibroids, may have increased vascularity with large vessels on the uterine surface that rupture during the menses.
- Endometriosis can cause heavy painful periods and dyspareunia.
- Heavy bleeding can occur in the presence of an intrauterine device (IUD), usually a nonhormonal releasing IUD.

Endometrial polyps

- Abnormal growth containing glands, stroma and blood vessels projecting from the lining of the uterus (endometrium) that occupies spaces small or large enough to fill the uterine cavity.
- Found during both reproductive and postmenopausal phases of life
- Majority of polyps are located in the fundus
- Range in size from about 5 mm to as large as filling the whole uterine cavity
- Can be found in all age groups, however, most common between age 40 and 49

- Endometrial polyp is attached to the uterine surface by a narrow elongated pedicle - pedunculated,
- Endometrial polyp have a large flat base, absence of a stalk – sessile
- Gross - smooth, spherical or cylindrical in structure and is tan to yellow in colour.
- Endometrium varies from normal cycling endometrium to simple or complex hyperplasia
- Usually benign lesions



- Pathophysiology

- Several theories proposed relating to the aetiology and pathogenesis of these lesions
 - Oestrogen stimulation (increased concentration of oestrogen receptors and a decreased expression of progesterone receptors)
 - Imbalance between mitotic activity and apoptosis in endometrium
 - Chromosomal rearrangements (translocation) in the stromal cells
 - Increased endogenous oestrogen and exogenous oestrogen administration

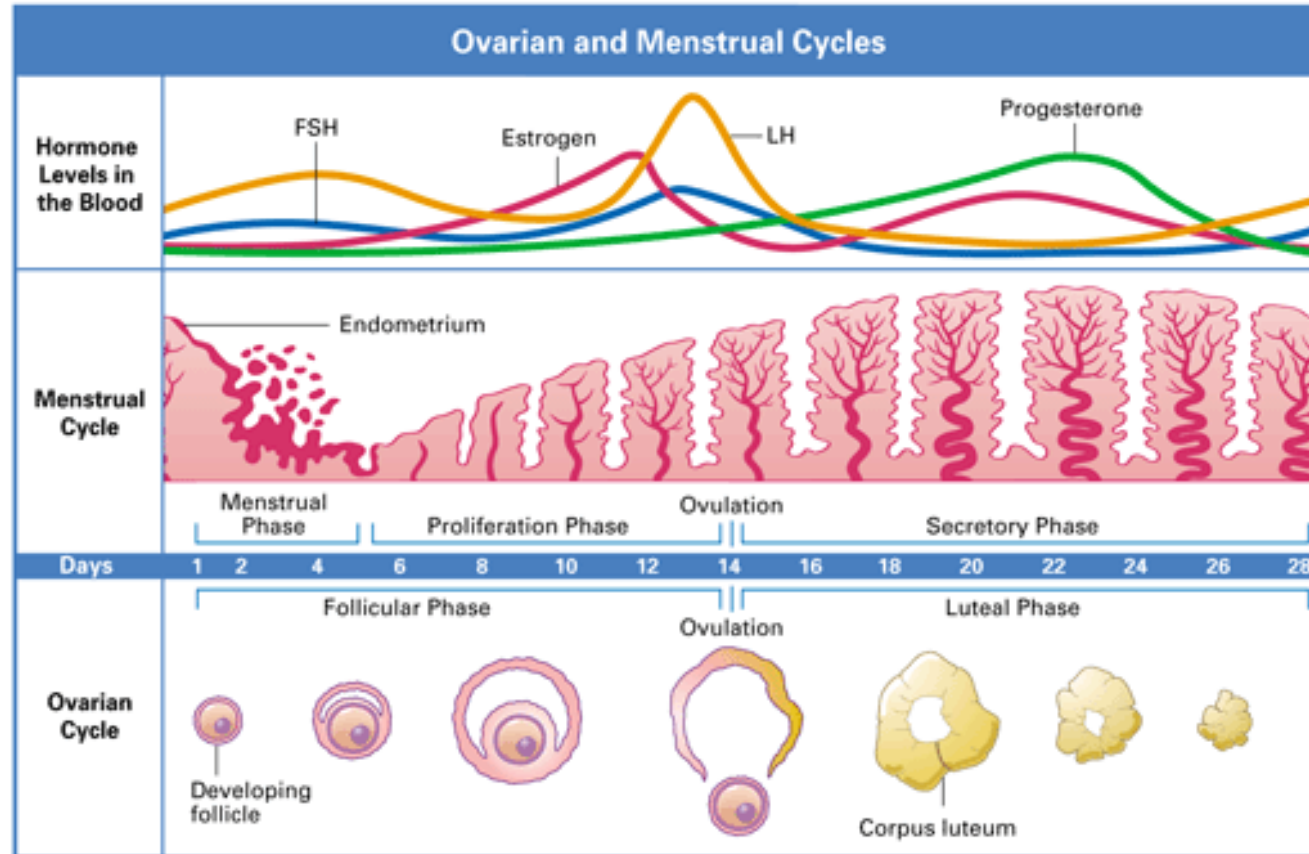
- **Clinical features**

- **Bleeding** - mostly asymptomatic lesions, although they can present with abnormal uterine bleeding due to stromal congestion within the polyp leading to venous stasis and apical necrosis
- **Infertility** – Cause implantation failure by *mechanical obstruction, by producing an inflammatory endometrial response, biochemical effects*; increased levels of matrix metalloproteinases and cytokines such as interferon gamma (toxic effects on sperm and inhibitory consequences on embryonic development), and glycodeclin (obstruct sperm-oocyte interaction) and low placenta protein 14 (immune suppressor favouring acceptance of the allogenic embryo)
- **Malignancy** - A very small fraction of polyps may become hyperplastic or show malignant transformation.

Endometrial hyperplasia

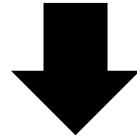
- Disordered proliferation of endometrial glands resulting from the unopposed estrogenic stimulation of the endometrial tissue with a relative deficiency of the counterbalancing effects of progesterone
- Oestrogen in excess of progesterone >>>> Irregular growth of the endometrium results in an abnormal gland-to-stroma ratio and presents in a continuum of the spectrum of changes in the endometrium (varying degrees of histopathological complexity and atypical features in the cells and nuclei) >>> may develop into endometrial cancer

Normal uterine cycle



- In the normal menstrual cycle, estrogen leads to proliferative endometrium. Normal endometrial tissue shows no glandular crowding, and the ratio of glands to stroma is less than 50%.
- After ovulation in the luteal phase, endometrium shows secretory changes under the effect of progesterone. Normal endometrial glands may show features such as minimal crowding and a small increase in gland-to-stroma ratio. Despite these features, the endometrial glands are organized and cells of the glands that do not show mitoses.

- Pathophysiology
 - Typical causes for endogenous estrogen excess include anovulatory cycles (perimenopause, polycystic ovarian syndrome (PCOS)), obesity, and estrogen secreting ovarian tumors. The exogenous causes include unopposed estrogen therapy, hormone replacement therapy (HRT), and tamoxifen



- Estrogen predominance and relative progesterone insufficiency
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- Four main categories
 - Simple hyperplasia without atypia: the number of glands increased, but the regular architecture was not compromised; progression rate to cancer 1%.
 - Simple hyperplasia with atypia: simple hyperplasia with features of nuclear atypia(marked variation in nuclear shape and size and prominent nucleoli); progression rate to cancer 8%
 - Complex hyperplasia without atypia: glands become more crowded and irregular; progression rate to cancer 3%.
 - Complex hyperplasia with atypia: complex hyperplasia with atypical features, progression rate to cancer 29%

Fibroid

- Leiomyomas, uterine myomas, fibromyomas
- Discrete, rounded, firm, white to pale pink, benign myometrial tumours composed mostly of smooth muscle with varying amounts of fibrous connective tissues
- Pathophysiology
 - Unclear
 - Genetic predisposition, exposure of circulating oestrogens

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- Types
 - **Subserosal or subperitoneal** - originate in the myometrium and grow out toward the serosal surface of the uterus, lying beneath the peritoneum
 - **Intramural or interstitial** - located within the uterine wall of the myometrium and may distort the shape of the uterine cavity and surface
 - **Submucosal fibroids** - originate in the myometrium and grow toward the endometrial cavity, protruding into the uterine cavity that they tend to compress
 - **Cervical leiomyomas** - rare type

- **Clinical features**
 - **Bleeding from the uterus**
 - Most common symptom. It may either be during the menstrual periods when the patient will have heavy and prolonged menses (menometorrhagia) or it may manifest as light spotting before and after the menses.
 - Due to the development or dilatation of endometrial venules which increase the flow during cyclical sloughing or to the increase in size of the uterine cavity by the fibromyomas
 - **Pain** - due to degeneration, infarction or torsion of a uterine fibroid. Sensation of pelvic heaviness or fullness or a feeling of a mass in the pelvis is particularly characteristic of large tumours. These may press on nerves within the bony pelvis, creating pain that radiates to the back or lower extremities
 - **Pressure effects** - May either be anteriorly on the bladder, causing mainly frequent micturation, and urinary incontinence or posteriorly, fibroids may increase the rectal pressure or cause constipation or tenesmus

QUESTION

- Outline the pathophysiology of reproductive dysfunction in fibroid.

Cervical Cancer

- Cancer of the cervix predominantly caused by persistent Human Papilloma Virus (HPV) infections
- Third leading malignancy among women after breast and colorectal cancers worldwide
- 200 identified HPV types >>> 12 have been designated as carcinogenic; HPV-16 accounting for 50% and HPV-18 accounting for 10% of cervical cancer cases
- HPV infection is sexually transmitted; HPV infection is often contracted during adolescence and early adulthood and because the infection is asymptomatic it may take 10 to 15 years to manifest changes in the cervix
- Introduction of HPV vaccines >>> cervical cancer rates have decreased

- Risk factors - Behavioral contributors and infectious contributors
 - Age of first sexual intercourse - first sexual encounter at a younger age or proximity to menarche increasing risk
 - Parity - Age less than 18 years at full-term pregnancy and multiple pregnancies (>4 vaginal births) increase risk
 - Multiple partners – increase risk
 - Smoking -
 - Tobacco byproducts >>> destroy the DNA cells in the cervix >>> contribute to the progression of cervical cancer
 - Smoking >>> a compromised immune system >>> reduce immunity against HPV infection
 - Co-infections – HIV, Genital Herpes
 - Prolonged use of oral contraceptives
 - Cervix dysplasia

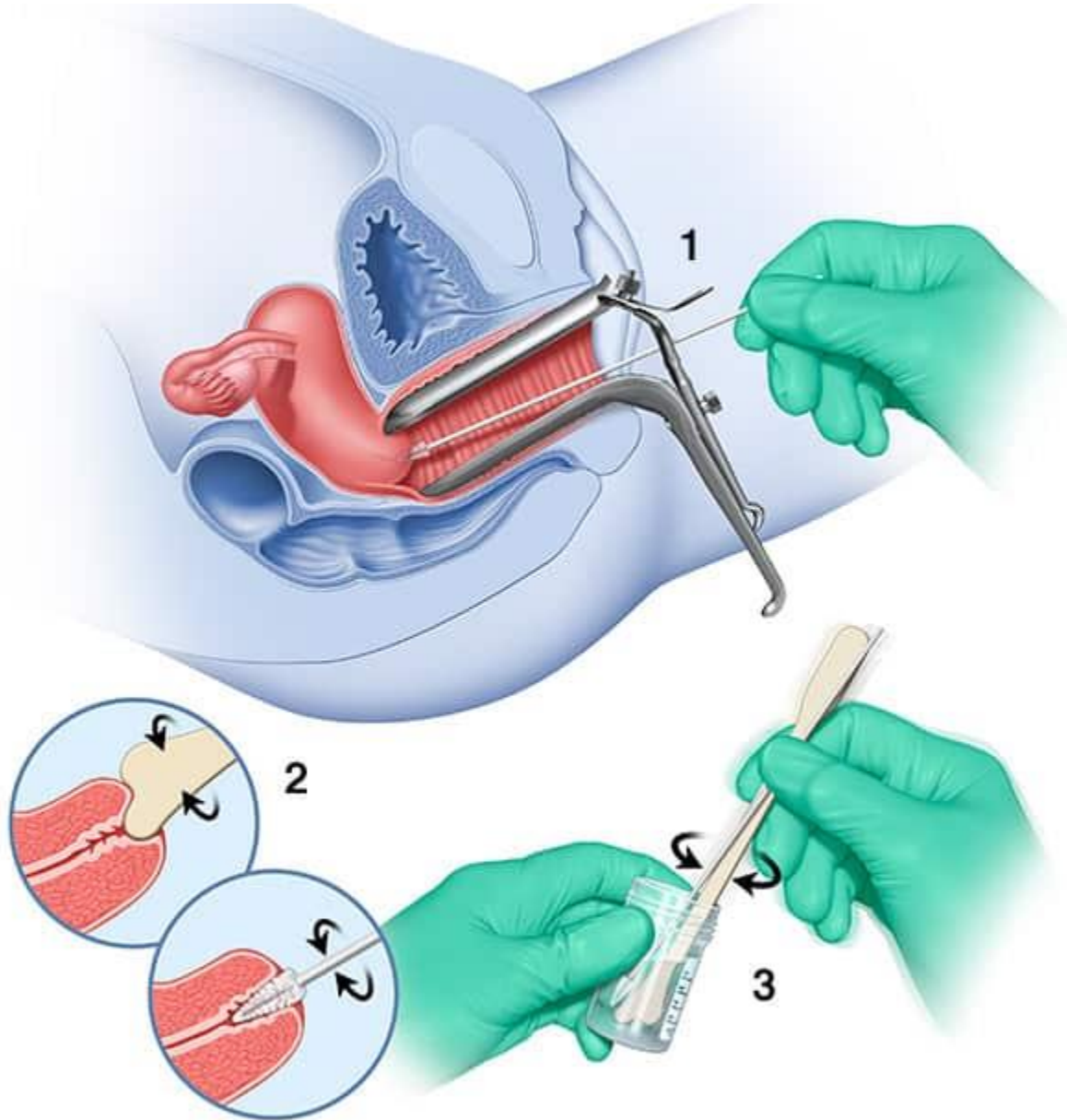
- Pathophysiology
 - Virus-specific factors and immune reactivity appear to be important
 - HPV replicative cycle is limited to stratified squamous epithelia. Most of the virus gene expression and replication take place in suprabasal cells that are undergoing terminal differentiation - a strategy that seems designed to evade immune surveillance
 - Viral genes, E6 and E7, which are the same two viral genes that are preferentially retained and expressed in cervical tumors >>> Keratinocyte immortalization >>> Ability to be passaged indefinitely is a characteristic of partial cellular transformation
 - Long interval between HPV infection and the development of cervical cancer suggests that factors other than viral infection are required for progression to high-grade dysplasia and tumor development

- Symptoms

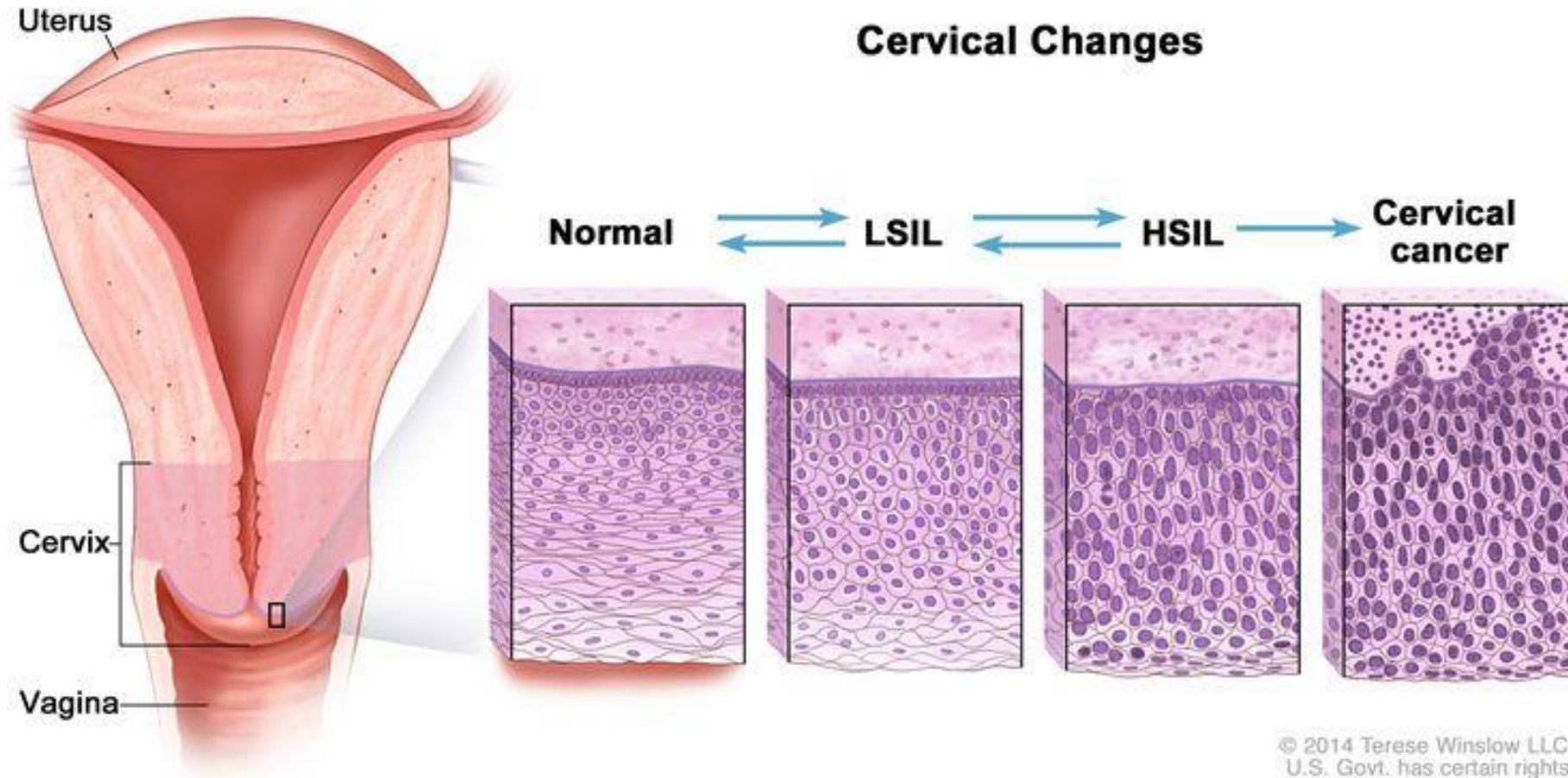
- Early-stage cervical cancer generally produces no signs or symptoms.
- Signs and symptoms of more-advanced cervical cancer include:
 - Vaginal bleeding after intercourse, between periods or after menopause
 - Watery, bloody vaginal discharge that may be heavy and have a foul odor
 - Pelvic pain or pain during intercourse

- Cervical cancer screening
 - Papanicolaou cytology (Pap) tests are the current standard for screening
 - Starting at age 30 >>> Pap test every 3 years, HPV co-testing every 5 years until age 65
 - Colposcopy is considered the definitive diagnostic test in the presence of abnormal examination or Pap test results
 - Cervical cancer is diagnosed based on the histologic evaluation of a cervical biopsy. The two most common histopathologic types of cervical cancer include squamous cell carcinoma (up to 85% of cases)^{15,35} and adenocarcinoma (up to 25%), including adenosquamous, and other histologies (6%)

Pap Smear



- Uses a vaginal speculum to hold your vaginal walls apart and to see the cervix.
- Sample of cells from your cervix is collected using a small cone-shaped brush and a tiny plastic spatula (1 and 2).
- Rinses the brush and spatula in a liquid-filled vial (3) and sends the vial to a laboratory for testing.



Cervical changes. The cervix is the lower, narrow end of the uterus that forms a canal between the uterus and vagina. Before cancer cells form in tissues of the cervix, the cells of the cervix go through abnormal changes called dysplasia. There are different types of dysplasia. Mild dysplasia, called low-grade intraepithelial lesion (LSIL) is one type. Moderate or severe dysplasia, called high-grade intraepithelial lesion (HSIL) is another type of dysplasia. LSIL and HSIL may or may not become cancer.

- Cervical Cancer Staging
 - Cervical cancer staging is the most prognostic factor, followed by nodal status, tumor volume, depth of cervical stromal invasion, and lymphovascular space invasion
 - Most types of cancer have stages I-IV. However, there are some types of cancer, including cervical cancer, which has stage 0 to IV
 - Stage 0: when abnormal cells are found in the inner lining of the cervix. Stage 0 is also called carcinoma in situ;
 - Stage I: when the cancer is confined in the cervix only;
 - Stage II: when the cancer has spread beyond the cervix but has not spread to the pelvic wall or to the lower third of the vagina;
 - Stage III: when the cancer has spread to the lower third of the vagina and/ or may have spread to the pelvic wall, and/or has caused kidney injury
 - Stage IV: when the cancer has spread to the bladder, rectum, or other parts of the body

PCOS and Infertility

- PCOS - Polycystic Ovary Syndrome
- Multifactorial disease
 - Genetic susceptibility – Genes involved in steroidogenesis and androgenic pathways
 - Environmental factors - obesity and insulin resistance
- Multiple conditions have been associated with PCOS, including infertility, metabolic syndrome, obesity, impaired glucose tolerance, DM-2, cardiovascular risk, depression, OSA, endometrial cancer, NAFLD/NASH.

PCOS - Epidemiology

- Most common endocrine pathology in reproductive-aged females worldwide
- Affecting about 5% and 15% of female of reproductive age - 4-20%, ??

PCOS – Clinical Manifestations

Menstrual irregularities - Oligo/amenorrhea ←

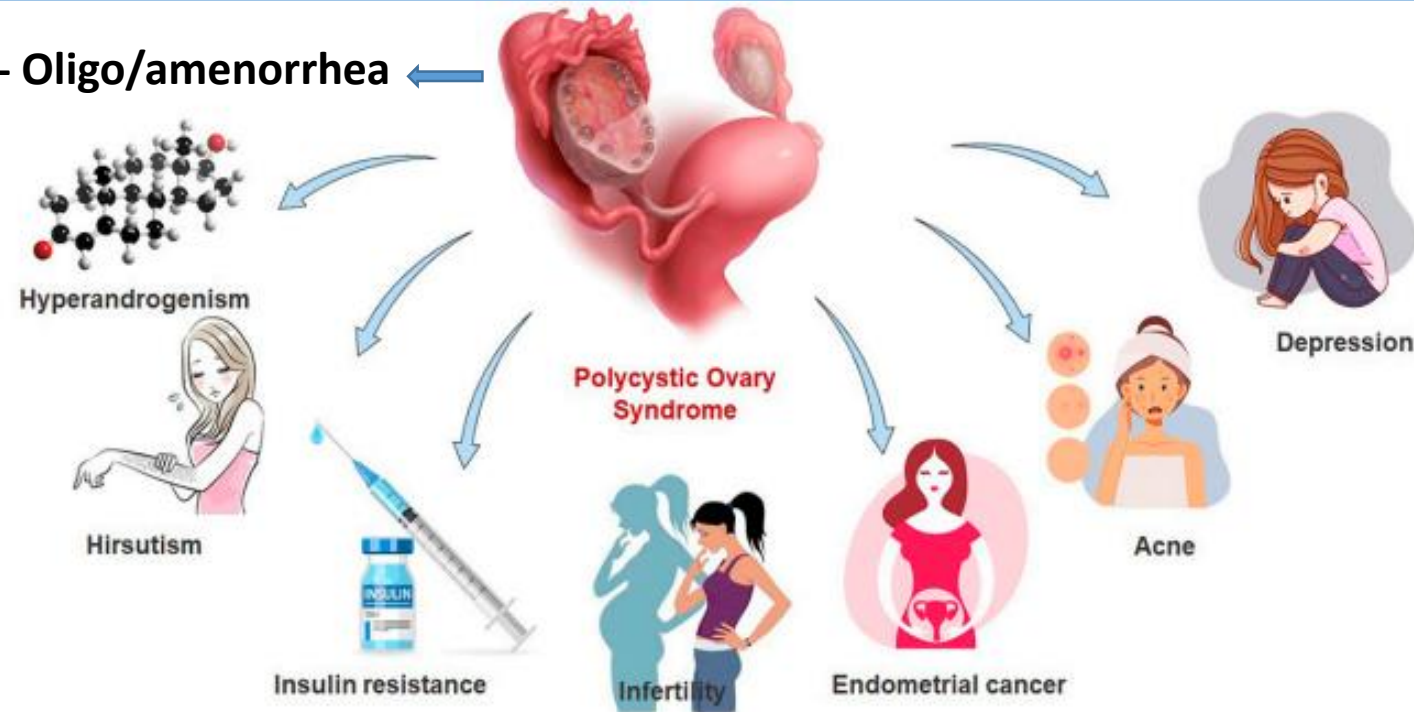


Figure 1. Clinical manifestations of polycystic ovary syndrome (PCOS). Complications including hyperandrogenism, infertility, hirsutism, insulin resistance, acne, and depression are markers for PCOS in women of reproductive age. Hyperandrogenism in PCOS identified by excess androgen levels in the circulation is associated with insulin-resistibility and visible features.

- Nearly all causes of PCOS are due to functional ovarian hyperandrogenism
 - Two third of cases - dysregulation of androgen secretion with an over-response of 17-hydroxyprogesterone (17-OHP) to gonadotropin stimulation
 - Remaining - atypical FOH lack of overresponse of 17-OHP, but testosterone elevation can detect it after suppressing adrenal androgen production
 - About 3% - isolated functional adrenal hyperandrogenism

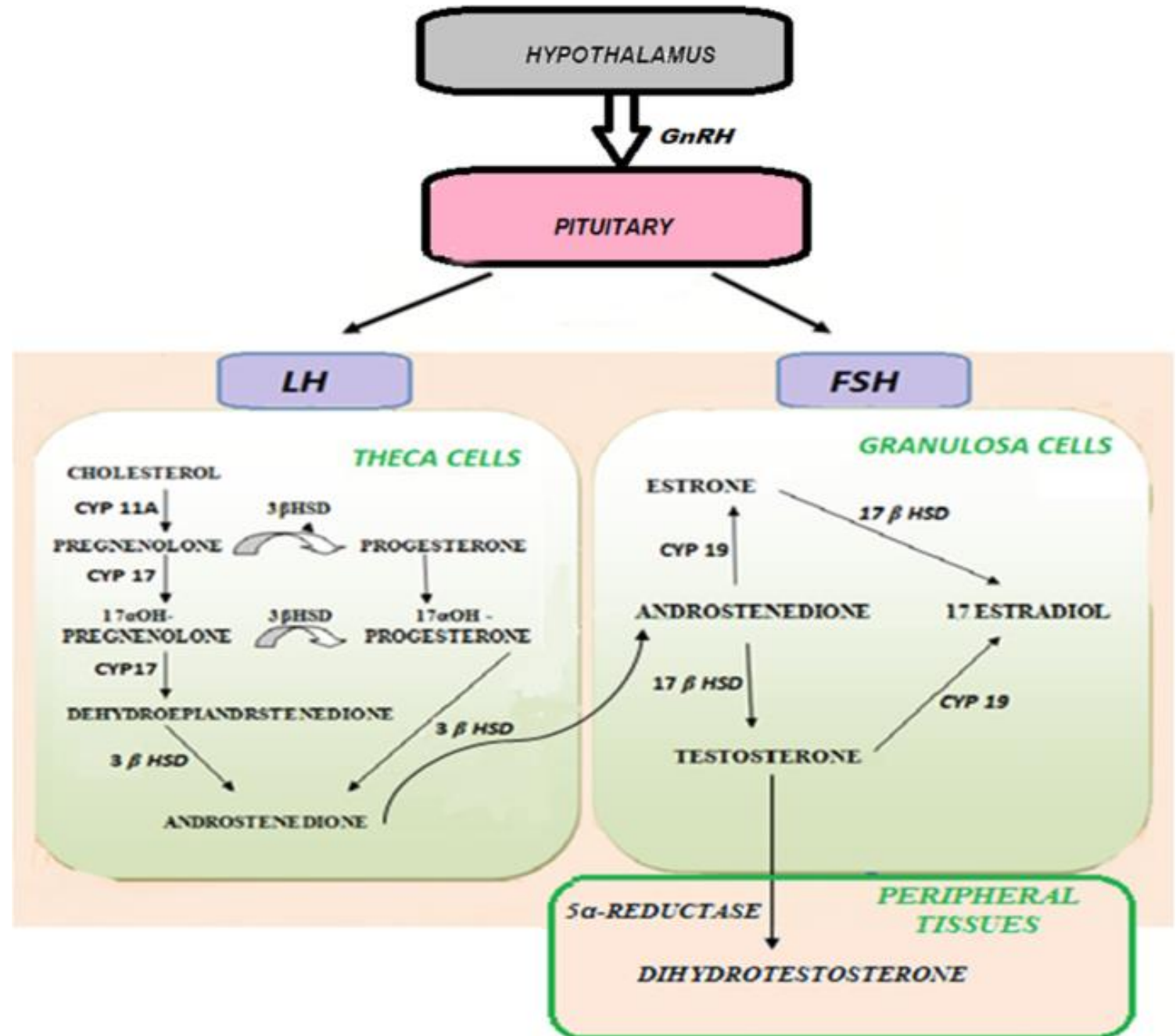
- Primary features
 - Hyperandrogenism
 - oligo anovulation
 - polycystic ovaries morphology
- Multifactorial- combination of heritable and environmental factors
 - Insulin excess >>> sensitize the ovary to luteinizing hormone (LH)
 - Theca >>> overexpress of most steroidogenic enzymes and proteins involved in androgen synthesis
 - Androgen and insulin excess >>> Granulosa cells prematurely luteinize primarily as a result of

- Under normal circumstances, the ovarian stroma provides a structural framework undergoing dynamic changes to support follicular growth. The developing oocyte and its surrounding scaffolding rely on endocrine, paracrine, and autocrine signaling mechanisms to maintain cell-to-cell communication and assure synchronized developmental progression
- Ovarian stroma from women with PCOS tends to be more rigid. Distorted interactions among the endocrine, paracrine, and autocrine factors responsible for likely contributes to the ovarian aspects of PCOS
- Ovarian dysregulation >>> PCOS

Hyperandrogenism in PCOS

- Androgen excess is regarded as the major driving force in the development of signs and symptoms of PCOS
- Excessive androgen production by ovaries as well as from adrenals contributes to hyperandrogenism
- Clinically, hyperandrogenism in women presents as hirsutism, acne, androgenic alopecia, and/or increased amounts of testosterone

Hypothalamic-pituitary-ovarian axis and steroidogenesis



Steroidogenesis in PCOS

- Genes that code for enzymes involved in the steroidogenic pathway are considered as candidates for PCOS.
 - Eg: *CYP11A* gene (cytochrome P450 side-chain cleavage enzyme gene), *CYP17* gene (cytochrome P450 17hydroxylase/17, 20-desmolase gene), and *CYP19* gene (aromatase)
- Ovarian theca cells of PCOS women >>> overexpress enzymes involved in androgen biosynthesis >>> increased production of 17-hydroxyprogesterone, testosterone, and androstenedione
- Decreased activity of aromatase enzyme >>> further increase androgens.
- Abnormalities in androgen production lead to hyperandrogenism in PCOS

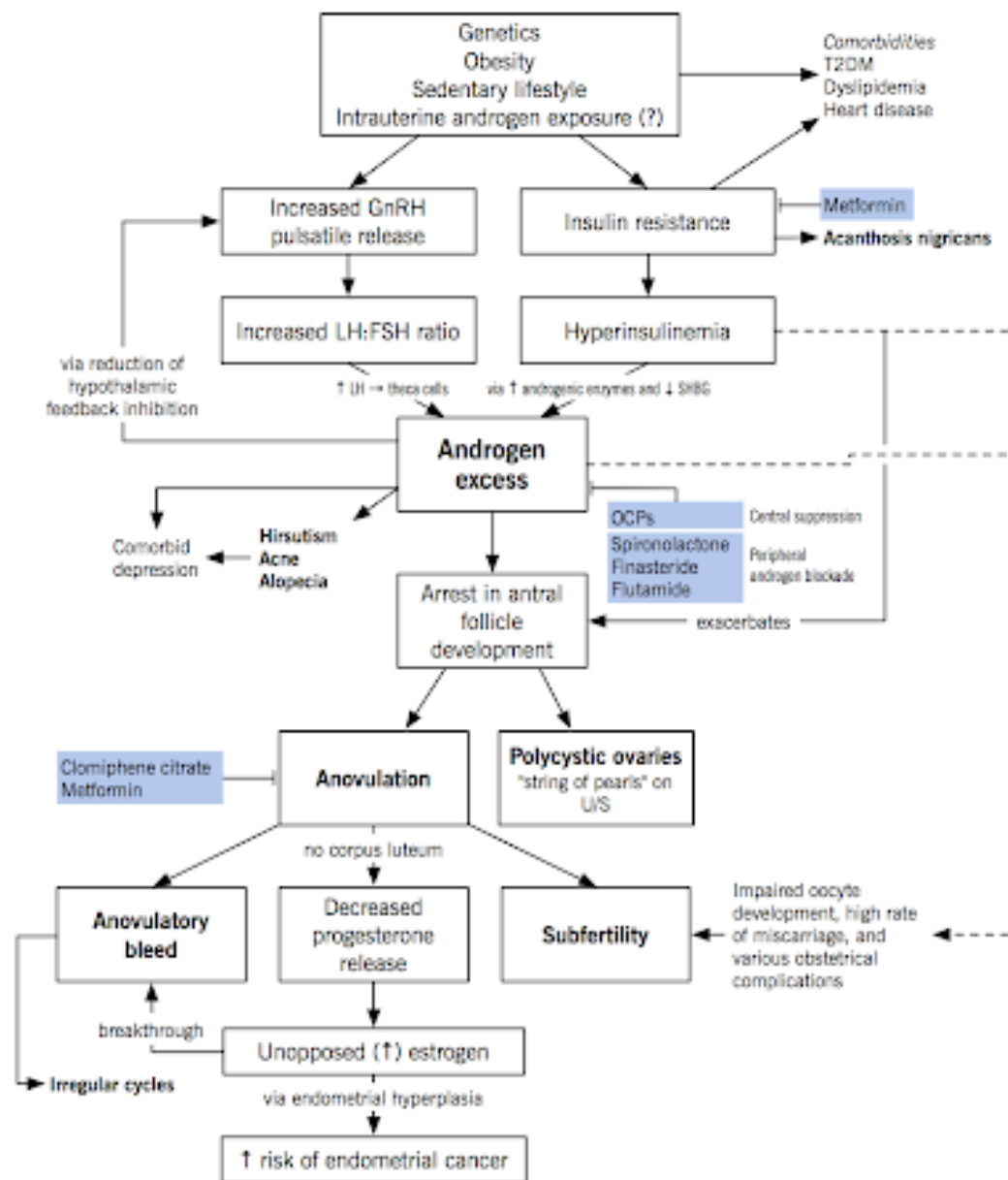
What happens in PCOS ?

- Dysregulation in the neuroendocrine system leads to an imbalance in the hypothalamic–pituitary–ovarian axis >>> Leading to the overproduction of gonadotrophins
 - Increased GnRH >>>> favors the production of LH over FSH >>> Elevated LH than FSH
 - Increased LH stimulation >>> numerous follicles in the theca cells of ovaries get arrested mostly in the preantral and antral stages >>> causing hyperplasia of theca cells >>> subsequent accumulation of follicular fluid >>> forming cyst-like structures along the periphery of the ovary (string of pearls-like appearance)

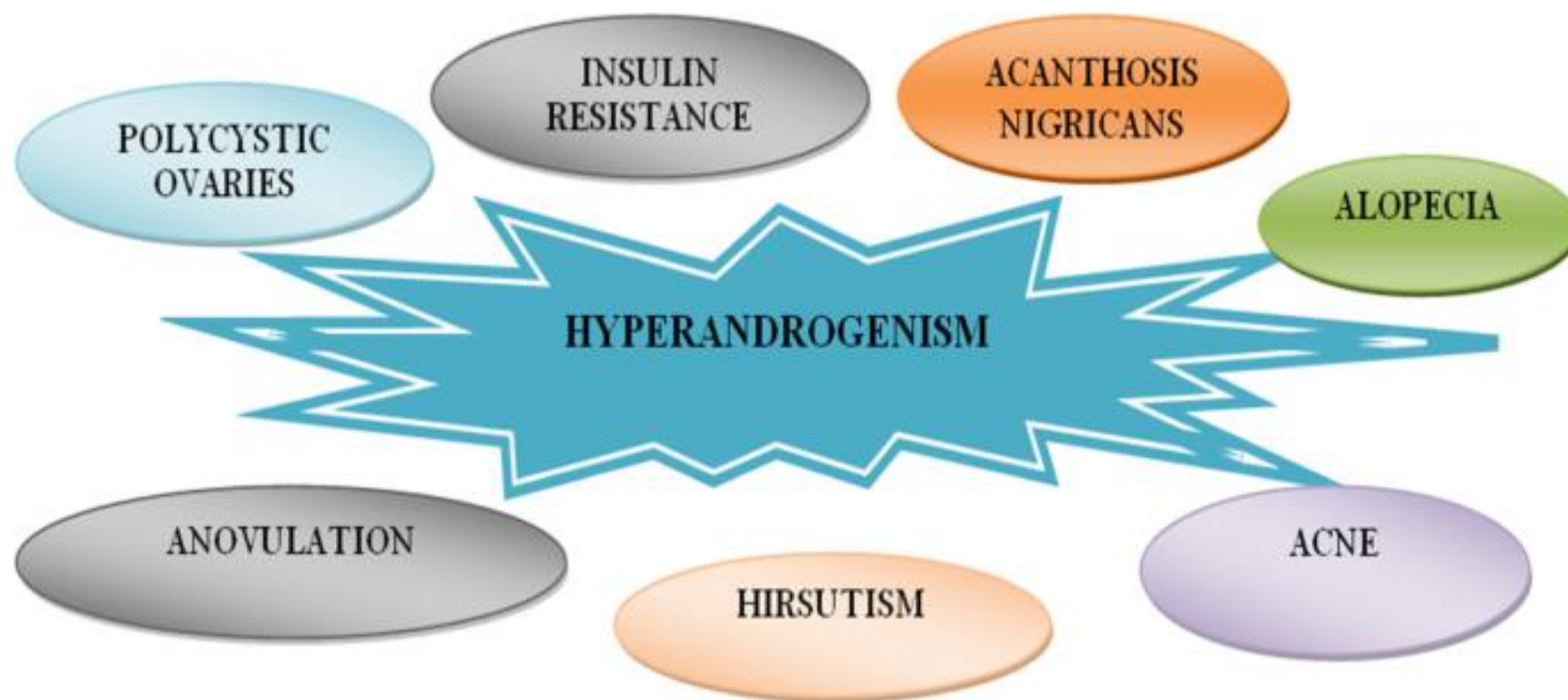
- Increased number of follicles + Increased expression of key enzymes involved in the androgen synthesis >>> produce an excessive amount of androgens
- Increased insulin secretion >>> mimics the tropic action of LH on ovarian theca cells >>> further causes an increase in androgens
- Biochemically, hyperandrogenism is defined as the high concentration of testosterone (T) and other calculated parameters of androgen excess like free testosterone (FT) and free androgen index (FAI).
 - 80% of testosterone is bound to sex hormone-binding globulin, 19% of it is bound to albumin, and only 1% circulates as a free testosterone
 - Measurement of testosterone and SHBG concentration, helps in calculating free androgen index which is more useful than measuring T alone

Pathophysiology of PCOS

Alex Robstein, Ragini Srinivasan, and Eric Wong



Clinical features in PCOS ?



Assessment of PCOS

- Most society guidelines have accepted that diagnosis of PCOS; most meet two out of three criteria: chronic anovulation, clinical or biological hyperandrogenism, and polycystic ovaries morphology in the absence of any other pathology - Rotterdam Criteria.
- The National Institute of Health criteria also requires clinical or biochemical hyperandrogenism and oligo or anovulation.
- The American Excess PCOS Society requires hyperandrogenism with one of two of the remaining criteria

Rationale for Treatment/ Management of PCOS

- **Lifestyle Modification**

- In overweight and obese PCOS women and adolescents >>> exercise and calorie-restrictive diets

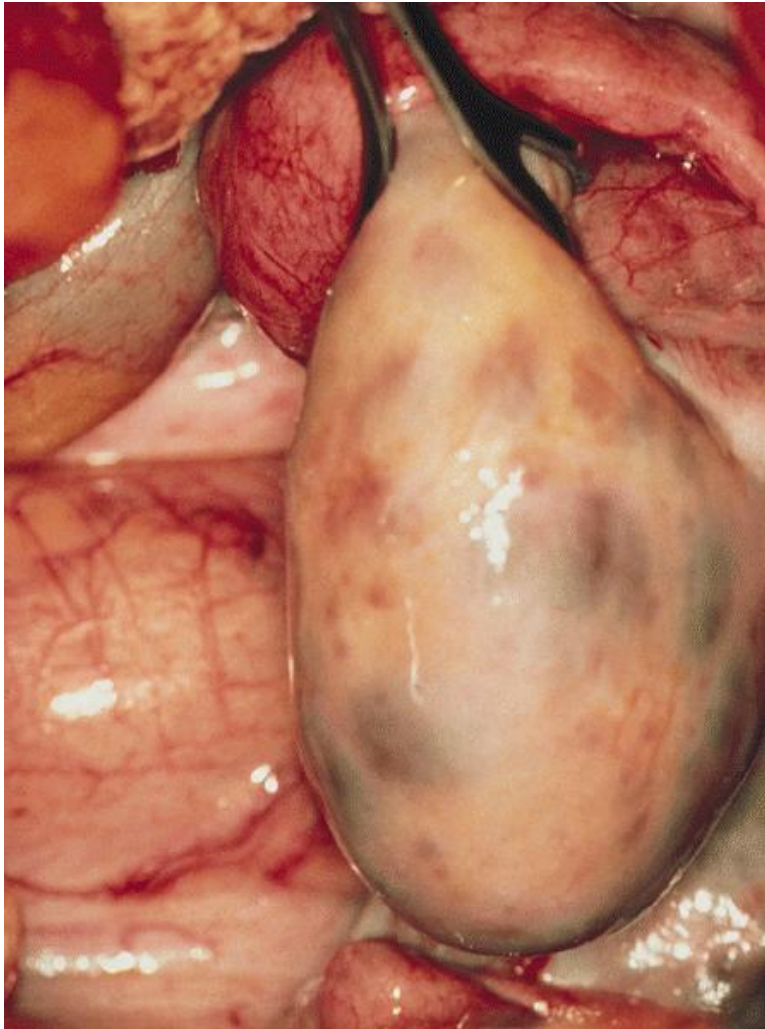
- **Hormonal Contraceptive**

- First-line treatment for menstrual abnormalities, hirsutism, and acne is a hormonal contraceptive, either oral contraceptive, patch, or vaginal rings
- Progestin component decreases LH levels, indirectly decreasing ovarian androgen production and increasing sex hormone-binding globulin
- Screening for contraindication for hormonal contraceptives – smoking, hypertension, DM

- **Metformin**
 - Starting metformin in PCOS patients with DM2 who fail lifestyle modifications
- **Infertility Treatment**
 - First-line therapy for infertility in PCOS patients is clomiphene citrate
 - Clomiphene >>> binds for a prolonged period to estrogen receptors in hypothalamus and depletes them, blocking the negative feedback inhibition effect of circulating endogenous estrogen >>> pulsatile release of a hypothalamic gonadotropin-releasing hormone (GnRH) >>> promoting the secretion of FSH and LH >>> indirectly stimulating ovulation
- **Treatment for Hyperandrogenism**



PCOS



Ectopic Pregnancy

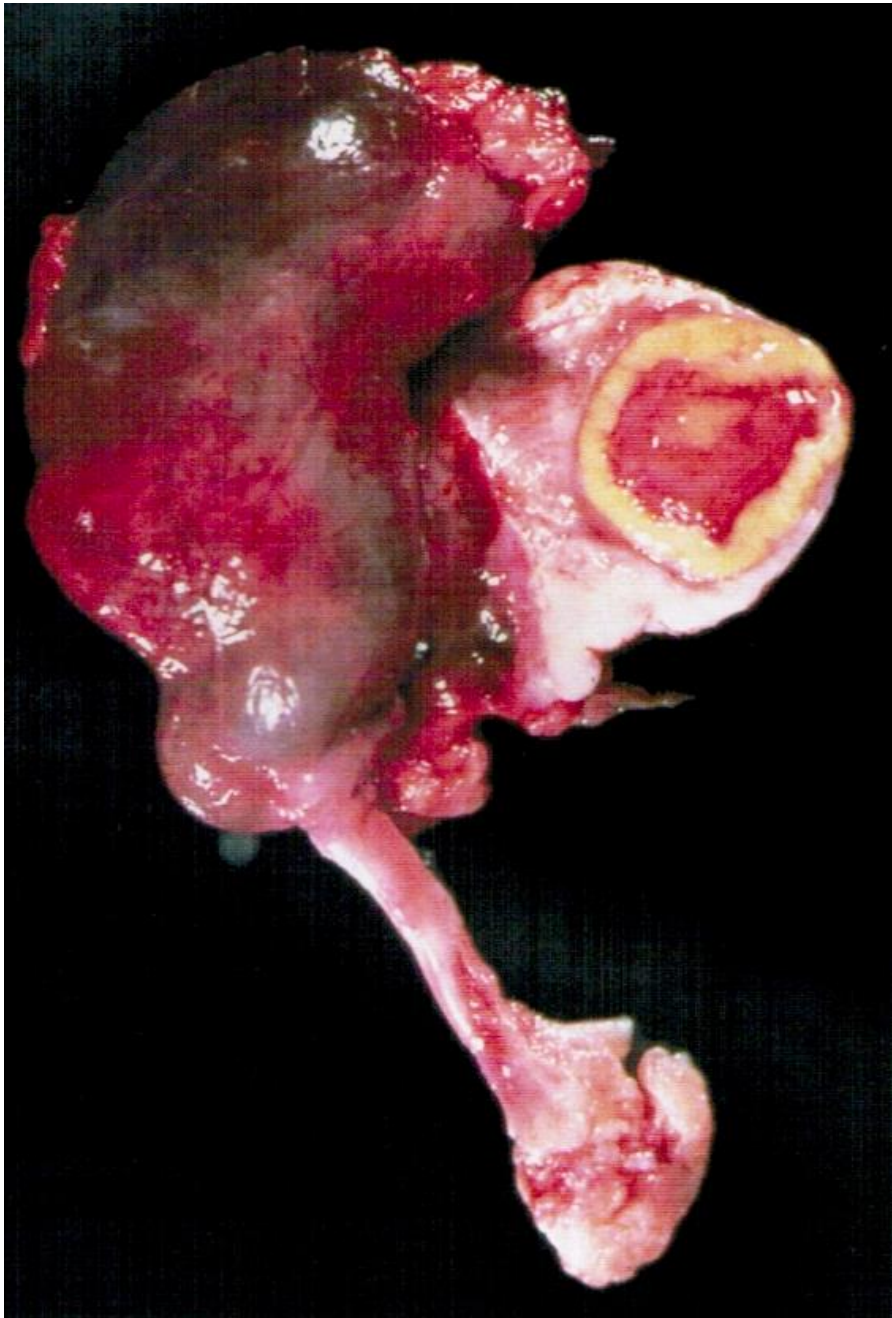
- Implantation of a pregnancy in a site other than the endometrial lining of the uterine cavity—ie, in the **fallopian tube, uterine cornua**, cervix, ovary, or abdominal or pelvic cavity.
- Life-threatening condition for a pregnant woman; such pregnancies cannot be carried to term and eventually rupture or involute.
- Early symptoms and signs include pelvic pain and vaginal bleeding. Hemorrhagic shock can occur with rupture
- Most ectopic pregnancies are located in the fallopian tube, and any history of infection or surgery that increases the risk of tubal adhesions or other abnormalities increases risk of ectopic pregnancy

Clinical features

- Most common clinical presentation is first trimester vaginal bleeding or abdominal pain
- Symptoms usually appear 6 - 8 weeks after the patient's last menstrual period but may occur even later, particularly if the pregnancy occurs outside of the fallopian tube
- No confirmed intrauterine pregnancy

Grossly Pathology

- Dilated tube with distended wall, with or without rupture
- Lumen contains blood clot, embryo, chorionic villi



The ampullary portion of the tube is markedly distended by blood. A corpus luteum of pregnancy is present in the adjacent ovary.



A portion of tubal lumen is distended with blood.

Outcome

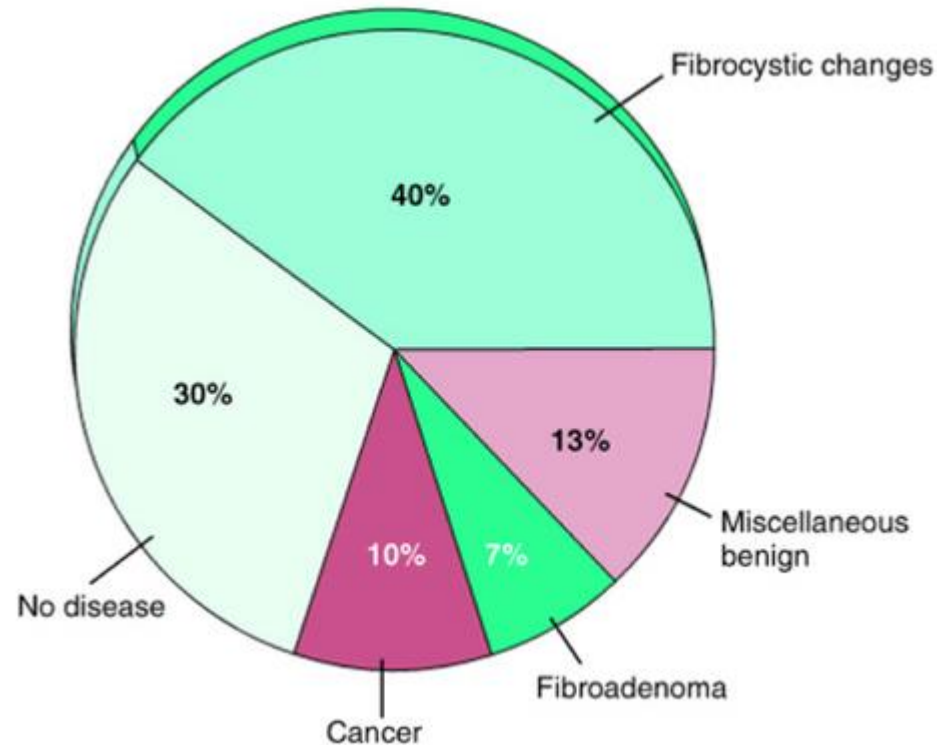
- As the pregnancy advances, tubal pregnancies may either diminish in size and spontaneously resolve, or increase in size and eventually lead to tubal rupture, with consequent maternal morbidity and mortality

Gestational trophoblastic disease

- Group of tumors defined by abnormal trophoblastic proliferation.
- There are 3 types of trophoblastic cells, cytotrophoblasts, syncytiotrophoblasts, and intermediate trophoblasts. The abnormal proliferation of these cells causes GTD.
- Transformations of cytotrophoblasts and syncytiotrophoblasts produce **hydatidiform moles and choriocarcinoma**.
- Intermediate trophoblasts are associated with **placental site trophoblastic tumor (PSTT)**, and **epithelioid trophoblastic tumor (ETT)**.

- **Hydatidiform moles**, also called molar pregnancies, are benign trophoblastic tumors comprising approximately 80% of all GTD.
- Molar pregnancies are caused by abnormal gametogenesis and/or fertilization. These pregnancies are divided into complete and partial moles.
- Complete molar pregnancy - occurs when an enucleated egg is fertilized either by two sperm or by a haploid sperm which then duplicates (46XX karyotype or 46XY karyotype)
- Partial molar pregnancies – occur as a result of the fertilization of a haploid ovum by two sperms or a haploid sperm duplicating upon fertilization of a haploid ovum (69XXX, 69XXY, or 69XYY).
- **Gestational trophoblastic neoplasias** are malignant placental tumors.

Breast lumps



- Cysts and Fibrosis – Nonproliferative change with increase fibrous stroma, dilation ducts and formation of cysts
- Epithelial hyperplasia – Proliferative lesions in ductules, ducts and lobules
- Inflammations – Acute mastitis – traumatic fat necrosis

Tumors of breast

Fibroadenoma

- Most common benign neoplasm
- Due to increase estrogen activity
- Gross – Discrete, solitary, freely movable nodule
- Enlarge late in menstrual cycle or during pregnancy, regress and calcify after menopause.

Breast carcinoma

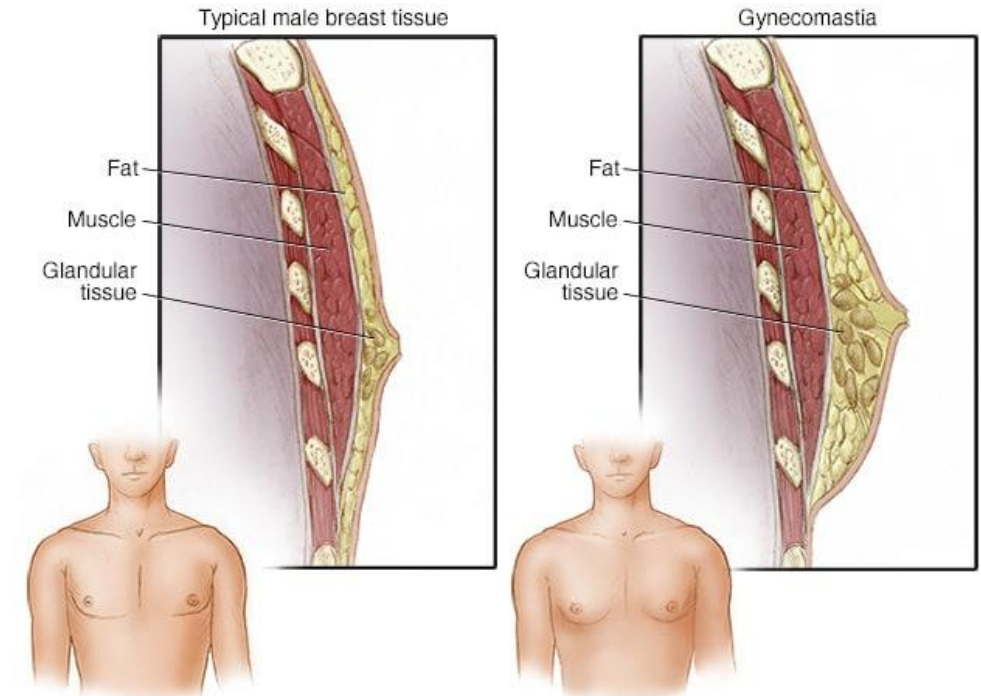
- Majority occur after 50 years
- Risk factors: Geographic variations, Age, Delayed child bearing, long duration between menarche and menopause, atypical proliferative lesions, family history(BRCA1 and BRCA2 genes), HRT, OCPs, Ionizing radiation of chest (before 30 years of age)
- Types
 - Noninvasive (Not penetrated the limiting basement membrane)
Eg: Ductal carcinoma in situ (Intraductal carcinoma), Lobular carcinoma in situ
 - Invasive (Infiltrative) – Eg: Invasive ductal carcinoma, Invasive lobular carcinoma, Medullary carcinoma, Colloid carcinoma, Tubular carcinoma, Other types

- Noninvasive Type
 - Arise from terminal duct lobular unit, Ductal carcinoma in situ tends to fill, distort and unfold lobular unit while Lobular carcinoma in situ usually expands but does not alter underlying lobular architecture
- Invasive Type
 - Invasive ductal carcinoma – Hard, palpable mass
 - Invasive lobular carcinoma – Palpable masses or Diffusely invasive pattern
- Spread of breast cancer
 - Lymphatic and hematogenous spread
- Clinical symptoms
 - Discrete, solitary, painless masses, 2-3cm size, regional lymph node (axillary) involvement
 - Mammography – can detect early

- Prognosis depend on tumor size, lymph node involvement, distant metastasis, tumor grade, histology type, proliferation rate, estrogen receptor status, aneuploidy (abnormal DNA content), overexpression of HER2/NEU (membrane bound protein)

Gynaecomastia

- A common condition that causes male breasts to swell and become larger than normal
- Most common in teenage boys and older men
- Signs vary from a small amount of extra tissue around the nipples to more prominent breasts. It can affect one or both breasts.
- Sometimes, the breast tissue can be tender or painful, but this isn't always the case.



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- Causes

- Hormone imbalance - imbalance between testosterone and estrogen
- Obesity – Increase estrogen, having excess fat
- Puberty
- Older age (>60 years) – less testosterone and more fat
- Testicular tumors
- End stage renal disease, Thyrotoxicosis
- Drug abuses

- Clinical symptoms

- Mostly bilateral, rubbery or firm, concentric around nipple, painful if recent onset or rapid enlargement

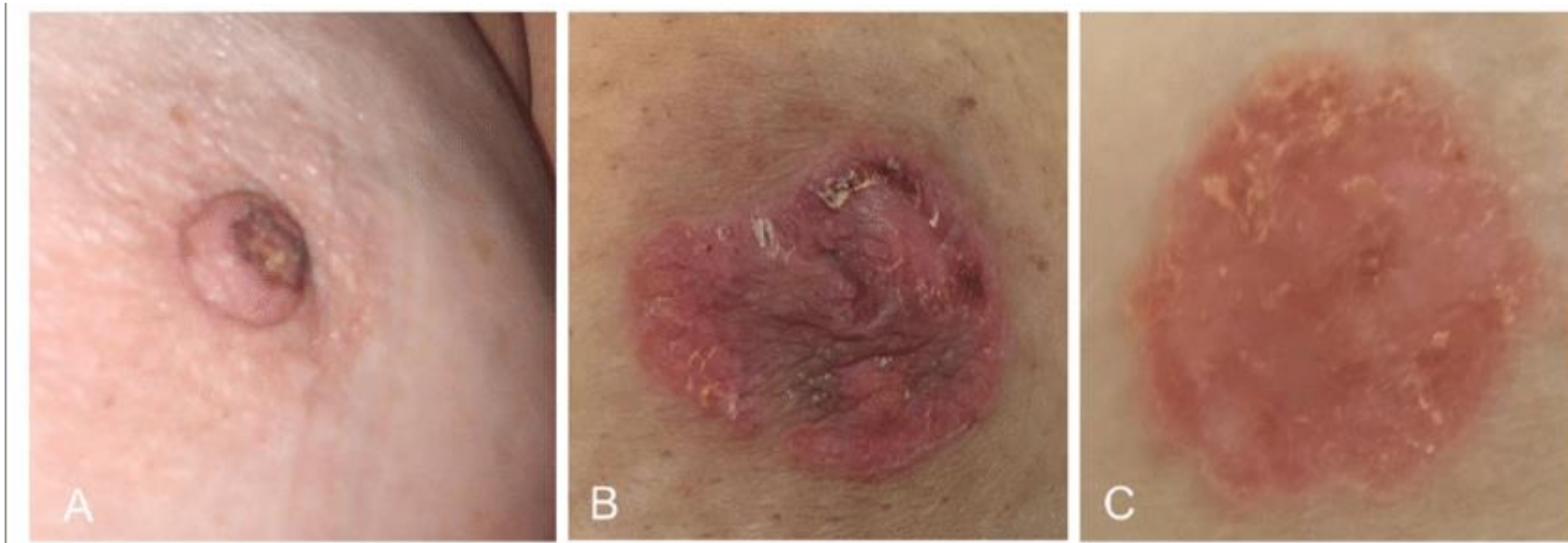
- Outcome

- Naturally resolve
- Medical/ Surgery

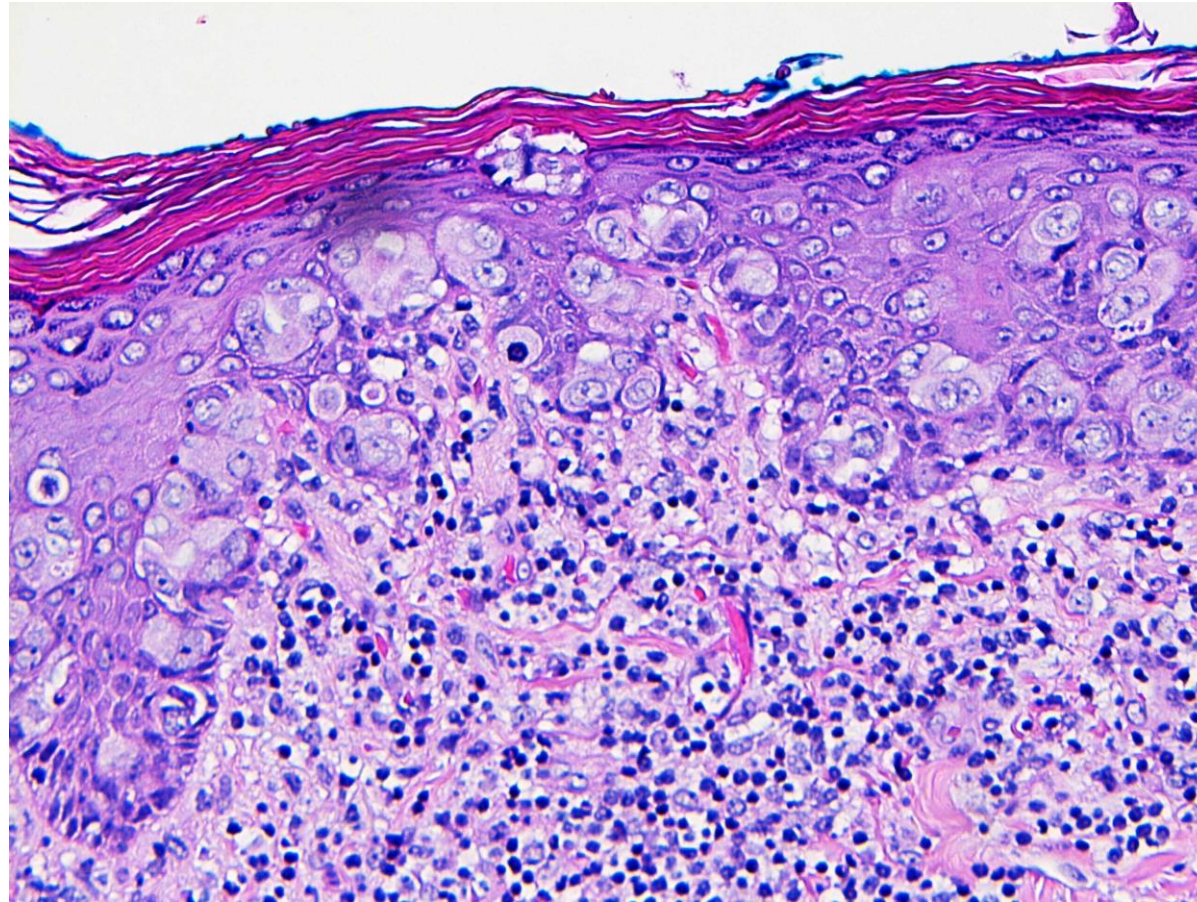
Paget's disease of breast

- A rare condition associated with breast cancer
- Causes eczema-like changes to the skin of the nipple and the area of darker skin surrounding the nipple (areola)
- Usually a sign of breast cancer in the tissue behind the nipple
- Is associated with some underlying breast malignancy, usually ductal carcinoma in situ (DCIS) or invasive ductal carcinoma
- Pathological features - eczematous ulcerative or vesicular lesions with clear yellowish exudate, Histology - presence of Paget cells, which are malignant, intraepithelial adenocarcinoma cells of variable sizes usually large in size, present singly or in the form of small groups within the epidermis of the nipple

- Clinical features - nipple discharge, pain, bleeding, burning, redness, scaling, and itching
- Outcome depends on the presence or absence of an underlying carcinoma



Cutaneous manifestations of Paget's disease of the nipple. a Crusted ulcerated papule in the center of the nipple. b Scaly erythematous plaque with erosions that destroys the nipple. c Scaly erythematous plaque that covers the entire nipple



Paget cells have abundant pale, clear cytoplasm, which often contains mucin. The nuclei are large and show prominent nucleoli. Paget cells are typically present at the basal portion of the epidermis but can extend throughout the entire thickness and be present very superficially.